

CoRoT-1b

R-2262

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-1_241127 · created 2026-08-16T01:29:16+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460641.9851 +/- 0.006 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.143 +/- 0.041
Transit depth (Rp/Rs)^2	2.05 +/- 1.18 [%]
Orbital Inclination (inc)	89.1 +/- 2.69
Airmass coefficient 1 (a1)	1.96 +/- 0.25
Airmass coefficient 2 (a2)	-0.56 +/- 0.13
Scatter in the residuals of the lightcurve fit is	12.38 %
Best Comparison Star	#3 - [449, 337]
Optimal Aperture	1.61
Optimal Annulus	13.0
Transit Duration (day)	0.0728 +/- 0.0067

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.143 $\pm$ 0.041 vs 0.1388 (deviation 0.05 σ)
Duration fit vs published	0.0728 d vs 0.1046 d (deviation 2.36 σ)
Mid-time O-C	4.4 min
Depth SNR	1.7
Residual scatter	12.38 %

Light curve

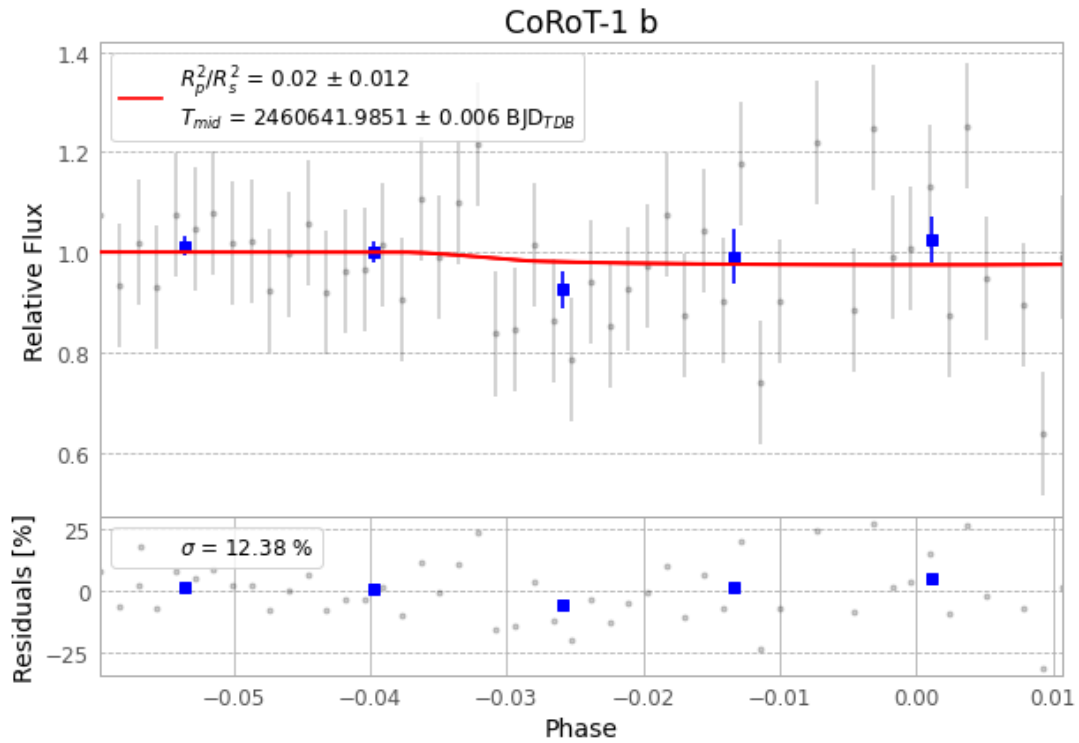


Figure 1 — FinalLightCurve_CoRoT-1 b_2024-11-27.png (SHA-256 3bf5f33383b13443...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [356, 255], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7f6a32ee094f909f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

52 FITS frames (34 MB), CoRoT-1241127092115.FITS → CoRoT-1241127115415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-1-241127.log (aedf0a484876...), FinalLightCurve_CoRoT-1 b_2024-11-27.png (3bf5f33383b1...), FinalLightCurve_CoRoT-1 b_2024-11-27.csv (57cce56fed0a...)

Compute resources

Wall time 118.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-16 01:29Z.